

HW 6 Question 6: Protein Analysis

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Question 6: Generalized Protein Analysis Function This section addresses the requirement to generalize the original analysis code into a reusable function for protein-drug interaction studies.

Documentation

Inputs: The function takes one input argument, `pdb_id`, which must be a character string representing a 4-letter Protein Data Bank (PDB) identifier (e.g., “4AKE”).

Purpose and Usage: The function automates the retrieval and processing of protein structural data to reduce manual errors.

Output: The function generates a line plot (B-factor vs. residue) that includes secondary structure information (SSE).

```
#' Analyze Protein B-Factors
#' @param pdb_id A 4-character PDB code string.
analyze_protein <- function(pdb_id) {

  # 1. Read the structural data for the specified protein [cite: 521]
  s <- read.pdb(pdb_id)

  # 2. Trim the protein structure to Chain A and Carbon Alpha (CA) atoms [cite: 522, 524]
  s.chain <- trim.pdb(s, chain = "A", elety = "CA")

  # 3. Extract the B-factor vector from the trimmed atom data [cite: 525, 526]
  s.b <- s.chain$atom$b

  # 4. Generate the plot with secondary structure (SSE) notations [cite: 527-532]
  plotb3(s.b, sse = s.chain, typ = "l", ylab = "Bfactor")
}
```

Example Output Below is the example output generated by calling the function for the “4AKE” kinase structure.

```
analyze_protein("4AKE")
```

```
## Note: Accessing on-line PDB file
```

